

Correlated Smith Visibility from NDF-Derived Shapes

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Abstract

Smith masking-shadowing is widely used in microfacet BRDFs to model the visibility in rough surfaces. Existing Smith models usually correlate visibility through height and or direction, but do not derive the directional correlation from the normal distribution function itself. In this work, we propose a first height-direction correlated Smith visibility term where the correlation is computed from an NDF-derived shape. For any isotropic NDF, we construct a representative shape whose normal distribution matches the target NDF. The NDF-dependent correlation term can then be recovered by measuring the overlap between the shadows cast by the NDF shape for the incoming and outgoing directions.

1. Introduction

Smith visibility is central to microfacet BRDFs, where it models masking and shadowing on rough microsurfaces. The separable Smith model treats the incoming and outgoing visibility events independently, while height-correlated variants couple them through a shared microsurface height. These models still miss part of the directional correlation where visibility attenuation should not be counted twice when the two directions are close.

We propose a Smith-based visibility function that makes height-direction correlation explicit through a new term $\Lambda_{\cap}(\omega_i, \omega_o)$. This term measures the shared attenuation between the incoming and outgoing directions as the overlap area of their shadows cast by a shape whose normal distribution matches the chosen NDF. For unit-roughness GGX, the corresponding shape is a hemisphere and the bidirectional overlap has a closed form solution. For any isotropic NDF, we instead construct a surface of revolution whose normals reproduce the target distribution, then use its shadow overlap to evaluate the same correlation term.

Section 2 reviews the Smith visibility models used for comparison. Section 3 introduces the overlap formulation and the NDF-derived geometry. Section 4 compares the resulting visibility terms.

2. Background

2.1. Microfacet BRDF

The microfacet BRDF models a rough surface as a distribution of randomly oriented infinitely small facets. The standard single-scattering BRDF for specular microfacets is

$$f_r(\omega_i, \omega_o) = \frac{F(\omega_i, \omega_m)D(\omega_m)G_2(\omega_i, \omega_o)}{4|n \cdot \omega_i||n \cdot \omega_o|}. \quad (1)$$

with n the surface normal and $\omega_m = \frac{\omega_i + \omega_o}{\|\omega_i + \omega_o\|}$ the halfway vector between ω_i and ω_o . The normal distribution function D gives the likelihood of microfacet normals, F is the Fresnel reflectance, and G_2 is the visibility function that accounts for masking and shadowing between the incident and outgoing directions. In this work, NDFs and visibility functions are *always* expressed in unit-roughness space. Anisotropic roughness can be recovered by stretching the microsurface [Hei14, AKDW22]. For a direction ω_α and normal $\omega_{m,\alpha}$ in a stretched space, the corresponding unit-roughness space vectors are

$$\omega = \frac{S_\alpha \omega_\alpha}{\|S_\alpha \omega_\alpha\|}, \quad \omega_m = \frac{S_\alpha^{-1} \omega_{m,\alpha}}{\|S_\alpha^{-1} \omega_{m,\alpha}\|}, \quad (2)$$

where $S_\alpha = \text{diag}(\alpha_x, \alpha_y, 1)$ is the roughness scaling matrix.

If D and G denote the unit-roughness NDF and visibility function, respectively, the corresponding rough functions are

$$D_\alpha(\omega_{m,\alpha}) = \frac{D(\omega_m)}{\alpha_x \alpha_y \|S_\alpha^{-1} \omega_{m,\alpha}\|^4}, \quad G_\alpha(\omega_{i,\alpha}, \omega_{o,\alpha}) = G(\omega_i, \omega_o). \quad (3)$$

2.2. Smith Visibility Models

In the Smith model, occlusion is derived from a microsurface whose facet orientations are statistically independent of their positions and heights. Smith visibility is often used in the separable form $G_2^{\text{sep}}(\omega_i, \omega_o) = G_1(\omega_i)G_1(\omega_o)$, which treats masking and shadowing events independently [WMLT07].

The unidirectional visibility is $G_1(\omega) = \frac{\chi^+(\omega \cdot n)}{1 + \Lambda(\omega)}$, with

$$1 + \Lambda(\omega) = \frac{1}{\cos \theta_\omega} \int_{\Omega} D(\omega_m) \langle \omega \cdot \omega_m \rangle d\omega_m, \quad (4)$$

where $\cos \theta_\omega = \omega \cdot n$.

Ross et al. [RDP05] partially addressed this limitation with a joint masking-shadowing term that accounts for the correlation induced by the shared height of the microsurface:

$$G_2^{\text{Ross}}(\omega_i, \omega_o) = \frac{\chi^+(\omega_i \cdot n) \chi^+(\omega_o \cdot n)}{1 + \Lambda(\omega_i) + \Lambda(\omega_o)}. \quad (5)$$

As shown in Fig. 1, accounting for this correlation increases reflectance at grazing angles.

Another source of correlation appears when the incoming and outgoing directions approach each other. Ashikhmin et al. [APS00] define bidirectional visibility as

$$G_2^{\text{Ash}}(\omega_i, \omega_o) = \lambda(\phi) G_1(\omega_i) G_1(\omega_o) + (1 - \lambda(\phi)) \min(G_1(\omega_i), G_1(\omega_o)), \quad (6)$$

where $\lambda(\phi) = \frac{4.41\phi}{4.41\phi + 1}$ is an empirical correlation factor with $\phi = \arccos(\omega_i \cdot \omega_o)$. As shown in Fig. 1, this model increases reflectance at grazing angles in retro-reflective configurations, where the incoming and outgoing directions are aligned.

Ginneken et al. [vSK98] combine height and directional correlation by defining the visibility function as

$$G_2^{\text{Gin}} = \frac{\chi^+(\omega_i \cdot n) \chi^+(\omega_o \cdot n)}{1 + \max(\Lambda_o, \Lambda_i) + \lambda(\omega_o, \omega_i) \min(\Lambda_o, \Lambda_i)}. \quad (7)$$

where $\Lambda_i = \Lambda(\omega_i)$ and $\Lambda_o = \Lambda(\omega_o)$. As shown in Fig. 1, this function inherits the behavior of the two previous models.

Visibility as Volume Attenuation. Because Smith assumes uncorrelated facets, Dupuy et al. [DHd16] interpret it as Beer-Lambert transmittance through a semi-infinite homogeneous microflake volume. In this view, a point at vertical depth t has optical thickness $\Lambda(\omega)t$ along direction ω . Averaging the transmittance over the height distribution gives

$$\int_0^{+\infty} \exp(-(1 + \Lambda(\omega))t) dt = \frac{1}{1 + \Lambda(\omega)}. \quad (8)$$

For two directions, sharing the same depth gives the height-correlated formulation

$$\int_0^{+\infty} \exp(-(1 + \Lambda(\omega_i) + \Lambda(\omega_o))t) dt = \frac{1}{1 + \Lambda(\omega_i) + \Lambda(\omega_o)}. \quad (9)$$

3. Height-direction correlated visibility

To also account for directional correlation, we propose to subtract from $\Lambda(\omega_i) + \Lambda(\omega_o)$ an intersection term $\Lambda_{\cap}(\omega_i, \omega_o)$ that represents the shared attenuation between the two directions:

$$\exp(-(1 + \Lambda(\omega_i) + \Lambda(\omega_o) - \Lambda_{\cap}(\omega_i, \omega_o))t), \quad (10)$$

which yields our height-direction correlated visibility formulation

$$G_2 = \frac{\chi^+(\omega_i \cdot n) \chi^+(\omega_o \cdot n)}{1 + \Lambda(\omega_i) + \Lambda(\omega_o) - \Lambda_{\cap}(\omega_i, \omega_o)}. \quad (11)$$

Note that Eq. 7 can be written in this new form using $\Lambda_{\cap}^{\text{Gin}} = (1 - \lambda(\omega_o, \omega_i)) \min(\Lambda_o, \Lambda_i)$.

3.1. GGX NDF visibility

Looking at Eq. 4, the Λ function can be interpreted as the projected shadow area of a surface with NDF D minus the surface base area. Since the GGX NDF is constant over the hemisphere, it therefore matches the normal distribution of a hemisphere. Looking at the GGX Smith Λ function:

$$\Lambda(\omega) = \frac{\sqrt{1 + \tan^2 \theta_\omega} - 1}{2} = \underbrace{\frac{1}{2 \cos \theta_\omega}}_{\text{Half ellipse}} - \underbrace{\frac{1}{2}}_{\text{Half disk}} \quad (12)$$

with $\cos \theta_\omega = \omega \cdot n$, we can see that the equation is equal to the projected shadow area of the hemisphere whose base has *unit* area. The shadow area is defined by the area of a half-ellipse minus a half-disk.

Following Lucas et al. [LRPB24], we define the directional correlation term $\Lambda_{\cap}$ as the area of the intersection between the incoming and outgoing projected shadow regions. As summarized in Fig. 2, the GGX overlap term is analytic because both the individual shadows and their intersections are simple conic regions. We refer the reader to Lucas et al. [LRPB24] for further details on the computation of shadow-intersection areas.

3.2. Extension to any isotropic unit-roughness NDFs

The GGX distribution is a special case where normals are generated by a simple hemisphere. This direct geometric analogy is lost for any isotropic NDF.

We thus propose to build this missing shape from the NDF itself. Given an isotropic NDF, we construct a surface of revolution with normals matching the NDF, then define $\Lambda_{\cap}(\omega_i, \omega_o)$ as the overlap of the two shadows cast by that shape. The surface of revolution makes the shape isotropic by design. Anisotropic variants are obtained afterward using the shape-invariance property.

3.2.1. Constructing the NDF Shape

Since we consider only isotropic NDFs, we seek a surface of revolution parameterized by $S(r, \phi) = [\sqrt{\pi}^{-1} r \cos \phi, \sqrt{\pi}^{-1} r \sin \phi, z(r)]$ with $r \in [0, 1], \phi \in [0, 2\pi]$ and $z(1) = 0$. The height profile $z(r)$ defines the shape height and thus normals. To recover $z(r)$ from an NDF, we need to match two quantities: the projected area at normal incidence of the shape inside radius r , and the cumulative distribution of the NDF up to $\theta(r)$. By construction, the projected area at normal incidence of the NDF shape is r^2 since the shape radius is $\sqrt{\pi}^{-1} r$. Matching this area with the cumulative NDF area gives

$$r^2 = C(\theta). \quad (13)$$

We use this relation by inversion: $\theta(r) = C^{-1}(r^2)$ gives the normal angle, and therefore the profile slope, at every radius. For the height profile, a normal tilted by $\theta(r)$ corresponds to a slope of magnitude $\tan(\theta(r))$. Since the height profile decreases from the center toward the boundary, this slope is negative:

$$dz = -\tan(\theta(r))dr. \quad (14)$$

We then recover the height profile by integrating these slopes. We start from height zero at the boundary, $z(1) = 0$, and integrate inward from radius 1 to radius r :

$$z(r) = \int_r^1 \tan(\theta(s)) ds. \quad (15)$$

Equations (14) and (15) are illustrated in Fig. 4.

Beckmann Example We now give an example using the unit-roughness Beckmann NDF. Its cumulative distribution function is

$$C_{\text{Beck}}(\theta) = 1 - \exp(-\tan^2 \theta), \quad (16)$$

from which Eq. (13) is defined as

$$\tan \theta(r) = \sqrt{-\ln(1 - r^2)}. \quad (17)$$

The resulting unit-roughness Beckmann profile is

$$z_{\text{Beck}}(r) = \int_r^1 \sqrt{-\ln(1 - s^2)} ds. \quad (18)$$

As shown in Fig. 4, the Beckmann shape is flatter than the GGX shape due to the long tail of GGX.

3.2.2. Validation

To validate our strategy of constructing a shape from the NDF, we verify that normals of the NDF-shape must reproduce the analytic NDF. Then we compute G_1 based on the shadow area cast by the NDF-derived shape which must reproduce the analytic Smith visibility. Validations are shown in Fig. 5.

4. Results

Figure 1 shows that our height correlation method mainly increases reflectance at grazing angle, while directional correlation increases reflectance around retro-reflective configurations giving similar results to Ginneken et al. Figure 3 compares our NDF-derived overlap against Ginneken et al. for GGX and Beckmann NDFs. Differences are only visible in retro-reflection for the GGX distribution. Due to the Beckmann NDF small intensities near retro-reflection, f_r values stay minimal when accounting for height-direction correlations. Height-direction correlation is therefore most useful for NDFs with pronounced backscattering behavior. For NDFs without a closed-form overlap, the intersection area of the reconstructed shape illustrated in Fig. 4 can be precomputed as a 3D lookup table over $(\theta_i, \theta_o, \phi_d)$.

5. Conclusion

We introduced a height-direction correlated Smith visibility model whose correlation term is the overlap between shadows cast from an NDF-derived shape. GGX admits a closed-form overlap, while other isotropic NDFs can use a precomputed representative shape. Our validation confirms that the construction preserves both the target NDF and G_1 . Evaluating the approach on Lambertian microfacet and multiple scattering BRDFs is left as future work. Visible normal distribution importance sampling may also benefit from NDF shapes as in current vNDF GGX importance sampling approaches.

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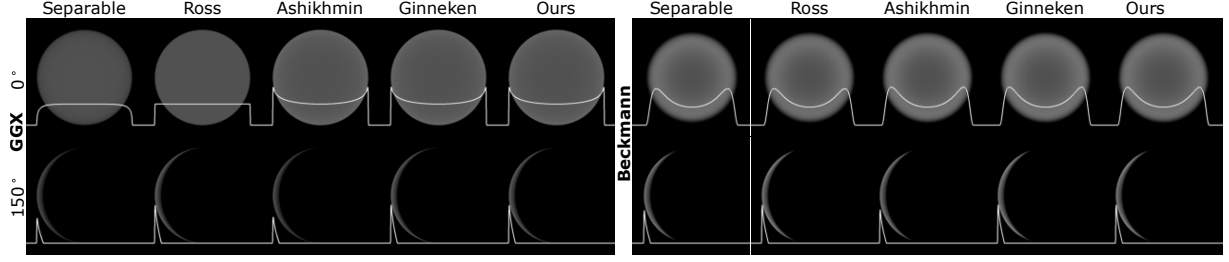


Figure 1: Render of a sphere using $F(\omega_i, \omega_o) = 1$ in retro-reflection (0°) and grazing-angle (150°) configurations, comparing separable Smith, Ross et al., Ashikhmin et al., Ginneken et al., and our height-direction correlated model for GGX and Beckmann distributions. The white curves show horizontal midline intensities.

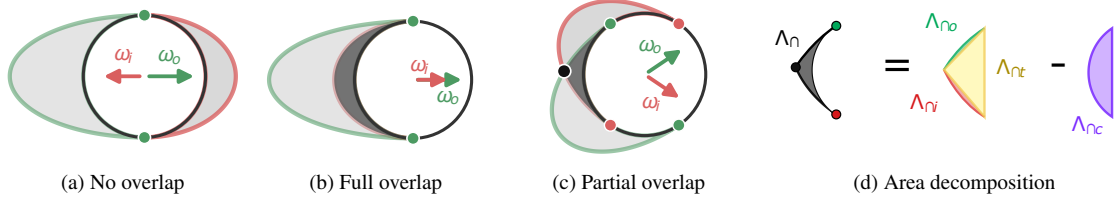


Figure 2: Computation of the GGX shadow-overlap area. The overlap term is $\Lambda_{\cap} = 0$ when the two projected shadows are disjoint (a). When the shadows are collinear and point to the same side, the smaller shadow is fully contained in the larger one, giving $\Lambda_{\cap} = \min(\Lambda(\omega_i), \Lambda(\omega_o))$ (b). In the partial case, the overlap decomposes as $\Lambda_{\cap} = \Lambda_{\cap t} + \Lambda_{\cap i} + \Lambda_{\cap o} - \Lambda_{\cap c}$, where $\Lambda_{\cap t}$ is a triangle area, $\Lambda_{\cap i}$ and $\Lambda_{\cap o}$ are ellipse-segment areas, and $\Lambda_{\cap c}$ is the circle-segment area. All terms are closed forms.

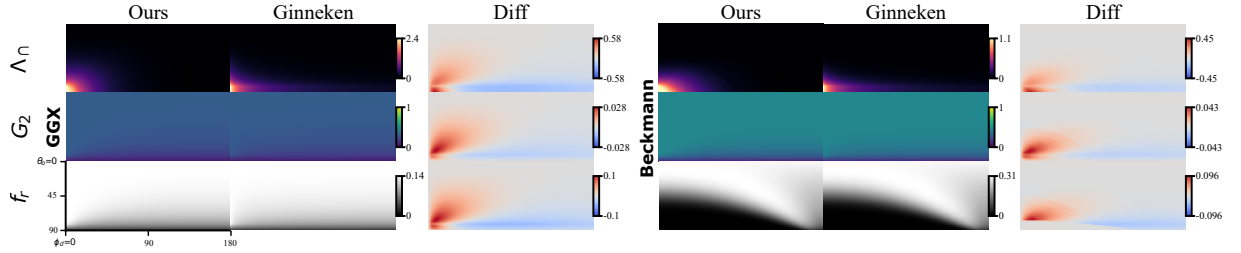


Figure 3: For GGX and Beckmann NDFs, we compare our model against Ginneken et al. and visualize the difference for $\Lambda_{\cap}$, G_2 , and the single-scattering BRDF f_r over outgoing directions with a fixed incident direction $\theta_i = 80^\circ$. Ginneken et al.'s $\Lambda_{\cap}$ has nearly the same shape for GGX and Beckmann, whereas our NDF-derived overlap changes with the distribution. The resulting change in G_2 only reduces attenuation near retroreflection. This also affects f_r near retroreflection, but it is visible only for GGX because its NDF has backscattering, unlike Beckmann.

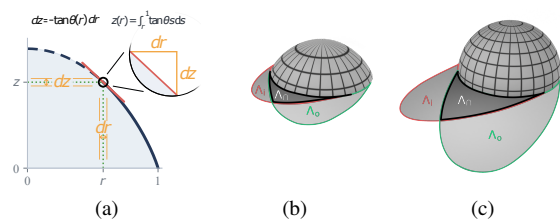


Figure 4: Construction of the shape height profile from an isotropic NDF (a). Shapes reconstructed from the Beckmann NDF (b) and the GGX NDF (c). The GGX shape matches a hemisphere.

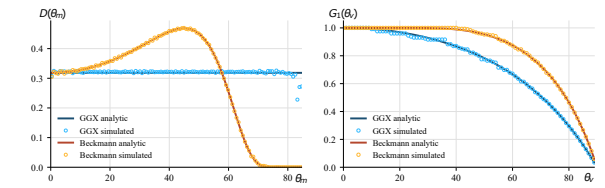


Figure 5: Validation of the NDF-to-shape construction for GGX and Beckmann. Shape normals reproduce the analytic NDFs (left), and shape shadow areas reproduce the analytic Smith G_1 (right).